In [147…

```python
import numpy as np


def cycle_matrix(n):
    D = np.diag([2] * n)

    A = np.zeros((n, n), dtype=int)
    for i in range(n):
        A[i, (i + 1) % n] = 1
        A[i, (i - 1) % n] = 1

    L = D - A
    return L

def wheel_matrix(n):
    D = np.diag([3] * n)

    A = np.zeros((n, n), dtype=int)
    for i in range(n-1):
        A[i, (i + 1) % (n-1)] = 1
        A[i, (i - 1) % (n-1)] = 1

    for i in range(n):
        A[i,n-1]=1
        A[n-1,i]=1


    L = D - A
    L[n-1,n-1] = 9

    return L


def list_eig(mat):
    eigenvalues, eigenvectors = np.linalg.eig(mat)

    print("Matrix:")
    print(mat)
    print("\nEigenvalues:")
    for i, val in enumerate(eigenvalues):
        print(f"λ{i+1}: {val:.4f}")
    print("\nEigenvecs:")
    for i, val in enumerate(eigenvectors):
        print(f"eigvec{i+1}: {val}")
    return eigenvectors



# if __name__ == "__main__":
#     list_eig(cycle_matrix(10))
#     # list_eigenvectors(cycle_matrix(10))
```

cycle

In [148…

```
cycle_eig = list_eig(cycle_matrix(10))
```

```
Matrix:
[[ 2 -1  0  0  0  0  0  0  0 -1]
 [-1  2 -1  0  0  0  0  0  0  0]
 [ 0 -1  2 -1  0  0  0  0  0  0]
 [ 0  0 -1  2 -1  0  0  0  0  0]
 [ 0  0  0 -1  2 -1  0  0  0  0]
 [ 0  0  0  0 -1  2 -1  0  0  0]
 [ 0  0  0  0  0 -1  2 -1  0  0]
 [ 0  0  0  0  0  0 -1  2 -1  0]
 [ 0  0  0  0  0  0  0 -1  2 -1]
 [-1  0  0  0  0  0  0  0 -1  2]]

Eigenvalues:
λ1: -0.0000
λ2: 0.3820
λ3: 1.3820
λ4: 2.6180
λ5: 4.0000
λ6: 3.6180
λ7: 3.6180
λ8: 0.3820
λ9: 1.3820
λ10: 2.6180

Eigenvecs:
eigvec1: [-0.31622777 -0.4472136  -0.4472136  -0.4472136   0.31622777 -0.4472136
  0.06480388  0.01396596 -0.00566843 -0.00933744]
eigvec2: [-0.31622777 -0.3618034  -0.1381966   0.1381966  -0.31622777  0.3618034
  0.20766368 -0.25143865  0.4235396  -0.42234726]
eigvec3: [-0.31622777 -0.1381966   0.3618034   0.3618034   0.31622777 -0.1381966
 -0.40081078 -0.42080224  0.26743029  0.2703624 ]
eigvec4: [-0.31622777  0.1381966   0.3618034  -0.3618034  -0.31622777 -0.1381966
  0.44086178 -0.42943368 -0.25825859  0.2552541 ]
eigvec5: [-0.31622777  0.3618034  -0.1381966  -0.1381966   0.31622777  0.3618034
 -0.31251857 -0.27403605 -0.42704288 -0.42811811]
eigvec6: [-0.31622777  0.4472136  -0.4472136   0.4472136  -0.31622777 -0.4472136
  0.06480388 -0.01396596 -0.00566843  0.00933744]
eigvec7: [-0.31622777  0.3618034  -0.1381966  -0.1381966   0.31622777  0.3618034
  0.20766368  0.25143865  0.4235396   0.42234726]
eigvec8: [-0.31622777  0.1381966   0.3618034  -0.3618034  -0.31622777 -0.1381966
 -0.40081078  0.42080224  0.26743029 -0.2703624 ]
eigvec9: [-0.31622777 -0.1381966   0.3618034   0.3618034   0.31622777 -0.1381966
  0.44086178  0.42943368 -0.25825859 -0.2552541 ]
eigvec10: [-0.31622777 -0.3618034  -0.1381966   0.1381966  -0.31622777  0.3618034
 -0.31251857  0.27403605 -0.42704288  0.42811811]
```

wheel

In [149…

```
wheel_eig = list_eig(wheel_matrix(10))
```

```
Matrix:
[[ 3 -1  0  0  0  0  0  0 -1 -1]
 [-1  3 -1  0  0  0  0  0  0 -1]
 [ 0 -1  3 -1  0  0  0  0  0 -1]
 [ 0  0 -1  3 -1  0  0  0  0 -1]
 [ 0  0  0 -1  3 -1  0  0  0 -1]
 [ 0  0  0  0 -1  3 -1  0  0 -1]
 [ 0  0  0  0  0 -1  3 -1  0 -1]
 [ 0  0  0  0  0  0 -1  3 -1 -1]
 [-1  0  0  0  0  0  0 -1  3 -1]
 [-1 -1 -1 -1 -1 -1 -1 -1 -1  9]]

Eigenvalues:
λ1: 10.0000
λ2: 0.0000
λ3: 1.4679
λ4: 2.6527
λ5: 4.8794
λ6: 4.0000
λ7: 4.8794
λ8: 4.0000
λ9: 1.4679
λ10: 2.6527

Eigenvecs:
eigvec1: [-0.10540926  0.31622777  0.47140452  0.47140452 -0.47140452  0.47140452
 -0.0594396  -0.06678194 -0.00477513 -0.07723751]
eigvec2: [-0.10540926  0.31622777  0.36111681  0.08185854  0.44297535 -0.23570226
 -0.10408807 -0.37073995  0.29933948 -0.47138121]
eigvec3: [-0.10540926  0.31622777  0.08185854 -0.44297535 -0.36111681 -0.23570226
  0.25506119  0.43752188  0.46338982 -0.08647147]
eigvec4: [-0.10540926  0.31622777 -0.23570226 -0.23570226  0.23570226  0.47140452
 -0.37527016 -0.06678194  0.41061491  0.44134998]
eigvec5: [-0.10540926  0.31622777 -0.44297535  0.36111681 -0.08185854 -0.23570226
  0.45021601 -0.37073995  0.16570872  0.23975071]
eigvec6: [-0.10540926  0.31622777 -0.44297535  0.36111681 -0.08185854 -0.23570226
 -0.47085917  0.43752188 -0.15673442 -0.35808543]
eigvec7: [-0.10540926  0.31622777 -0.23570226 -0.23570226  0.23570226  0.47140452
  0.43470976 -0.06678194 -0.40583978 -0.36411247]
eigvec8: [-0.10540926  0.31622777  0.08185854 -0.44297535 -0.36111681 -0.23570226
 -0.34612794 -0.37073995 -0.4650482   0.2316305 ]
eigvec9: [-0.10540926  0.31622777  0.36111681  0.08185854  0.44297535 -0.23570226
  0.21579798  0.43752188 -0.3066554   0.4445569 ]
eigvec10: [ 9.48683298e-01  3.16227766e-01  1.98803348e-16  1.51592507e-17
  1.76144701e-16 -1.29013861e-16  4.51293553e-17  2.29912603e-17
  1.42264122e-17  1.37648139e-17]
```

plots

In [150…

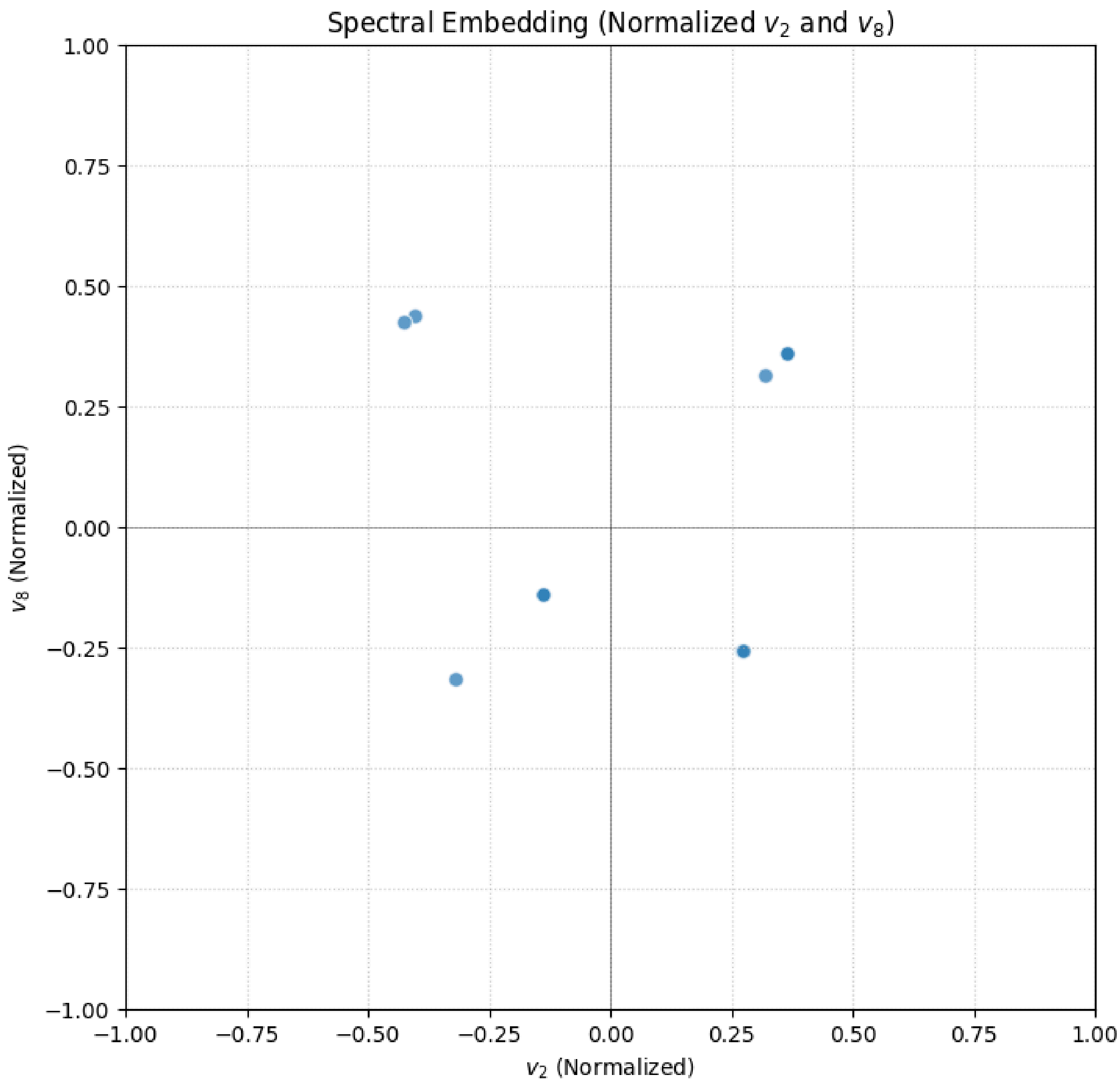
```python
import matplotlib.pyplot as plt


def plot_spectral_embedding(v2, v3,num1, num2, title=""):

    norm_v2 = np.linalg.norm(v2)
    norm_v3 = np.linalg.norm(v3)
```

```
    # Normalize each vector
    v2_normalized = v2 / norm_v2
    v3_normalized = v3 / norm_v3

    # Use these directly as coordinates
    x = v2_normalized
    y = v3_normalized

    plt.figure(figsize=(8, 8))
    plt.xlim(-1, 1)
    plt.ylim(-1, 1)
    plt.scatter(x, y, alpha=0.7, edgecolors='w', s=50)



    plt.axhline(0, color='black', linewidth=0.5, alpha=0.5)
    plt.axvline(0, color='black', linewidth=0.5, alpha=0.5)
    plt.title(f"Spectral Embedding (Normalized $v_{num1}$ and $v_{num2}$)")
    plt.xlabel(f"$v_{num1}$ (Normalized)")
    plt.ylabel(f"$v_{num2}$ (Normalized)")
    # plt.axis('equal') # Maintain aspect ratio
    plt.grid(True, linestyle=':', alpha=0.6)
    plt.show()
```

cycle min eigs

In [151…

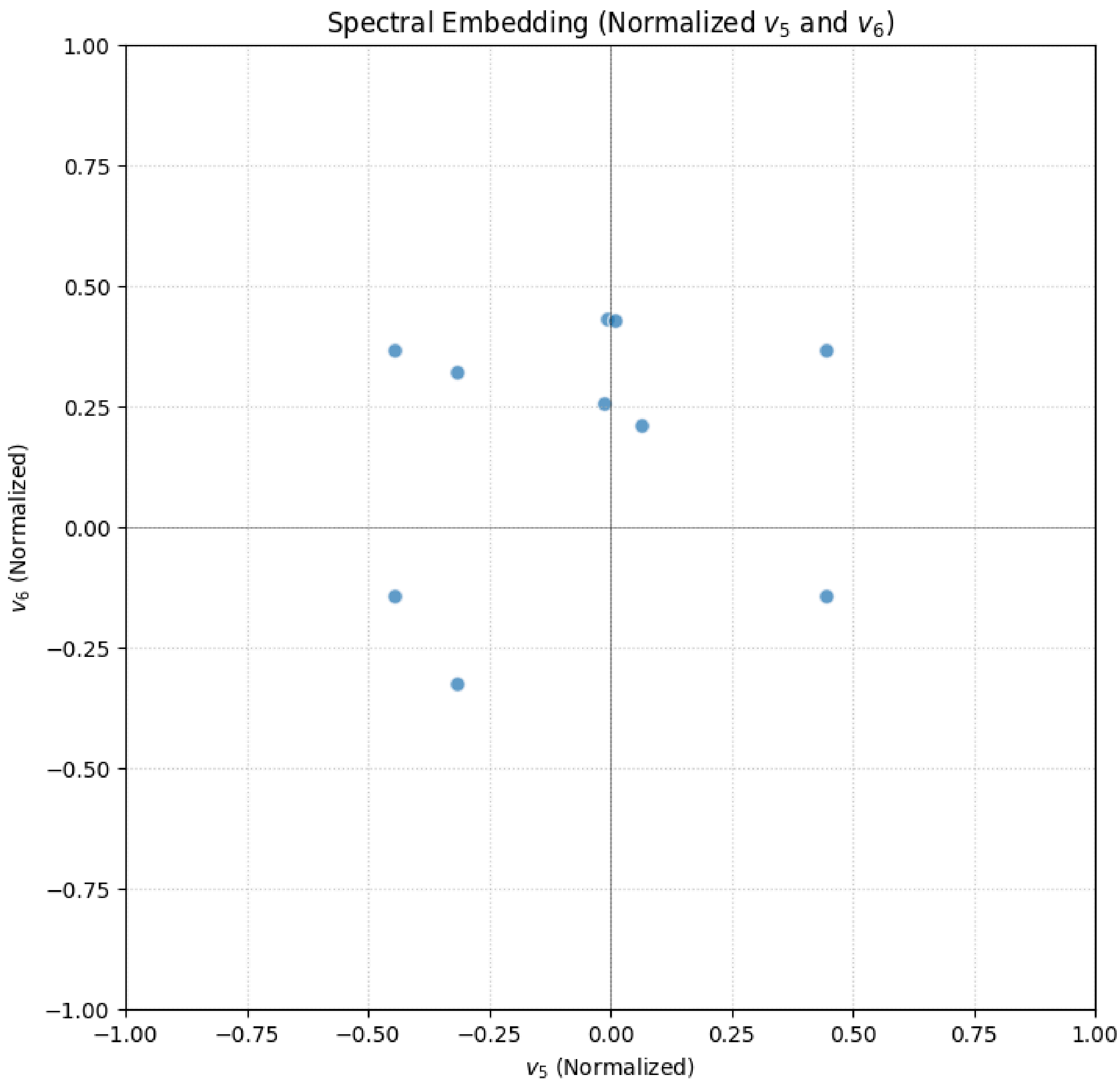
```
plot_spectral_embedding(cycle_eig[2],cycle_eig[8],2,8)
```



cycle max eigs

In [152…

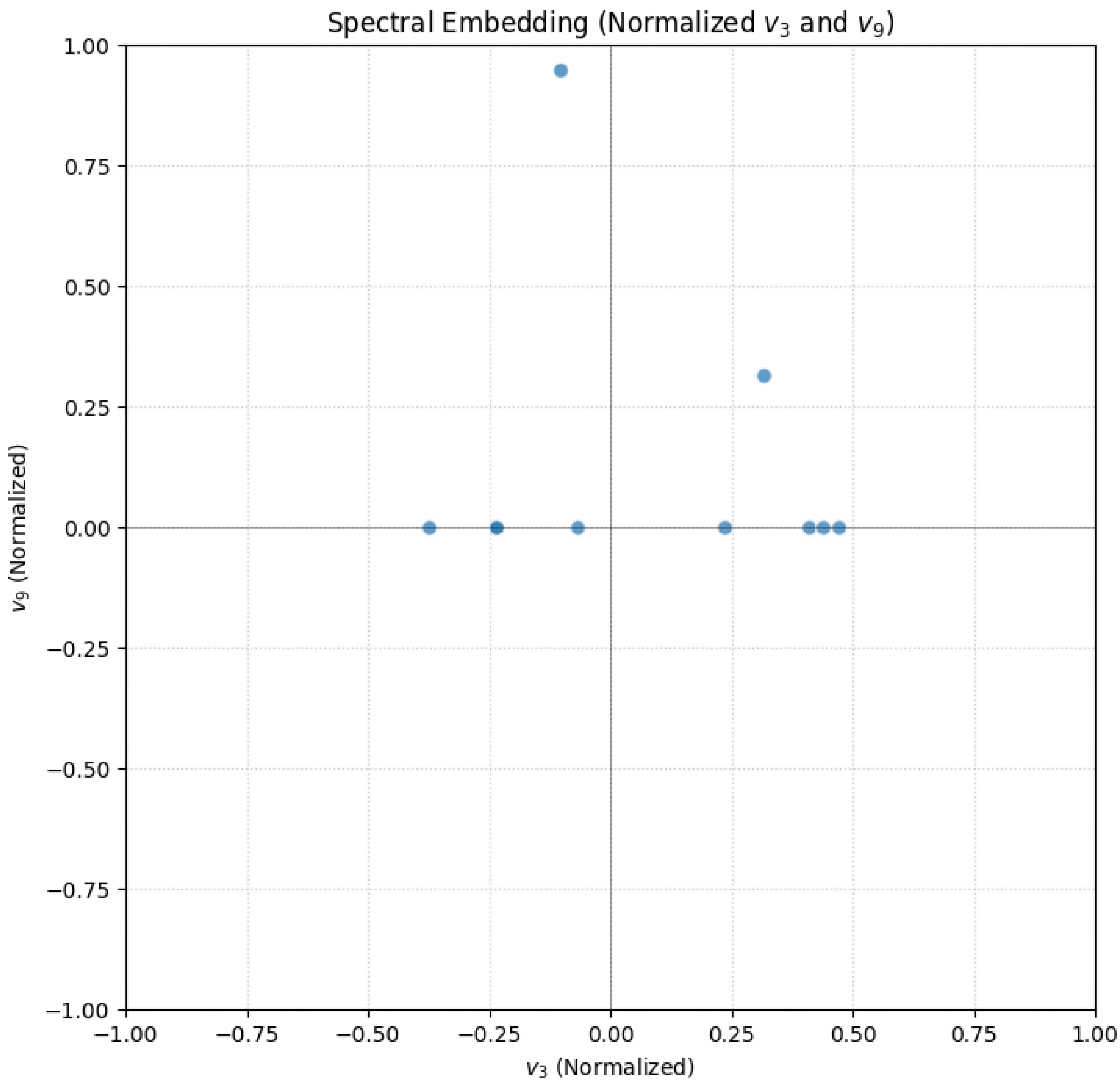
```
plot_spectral_embedding(cycle_eig[5],cycle_eig[6],5,6)
```



wheel min eigs

In [153…

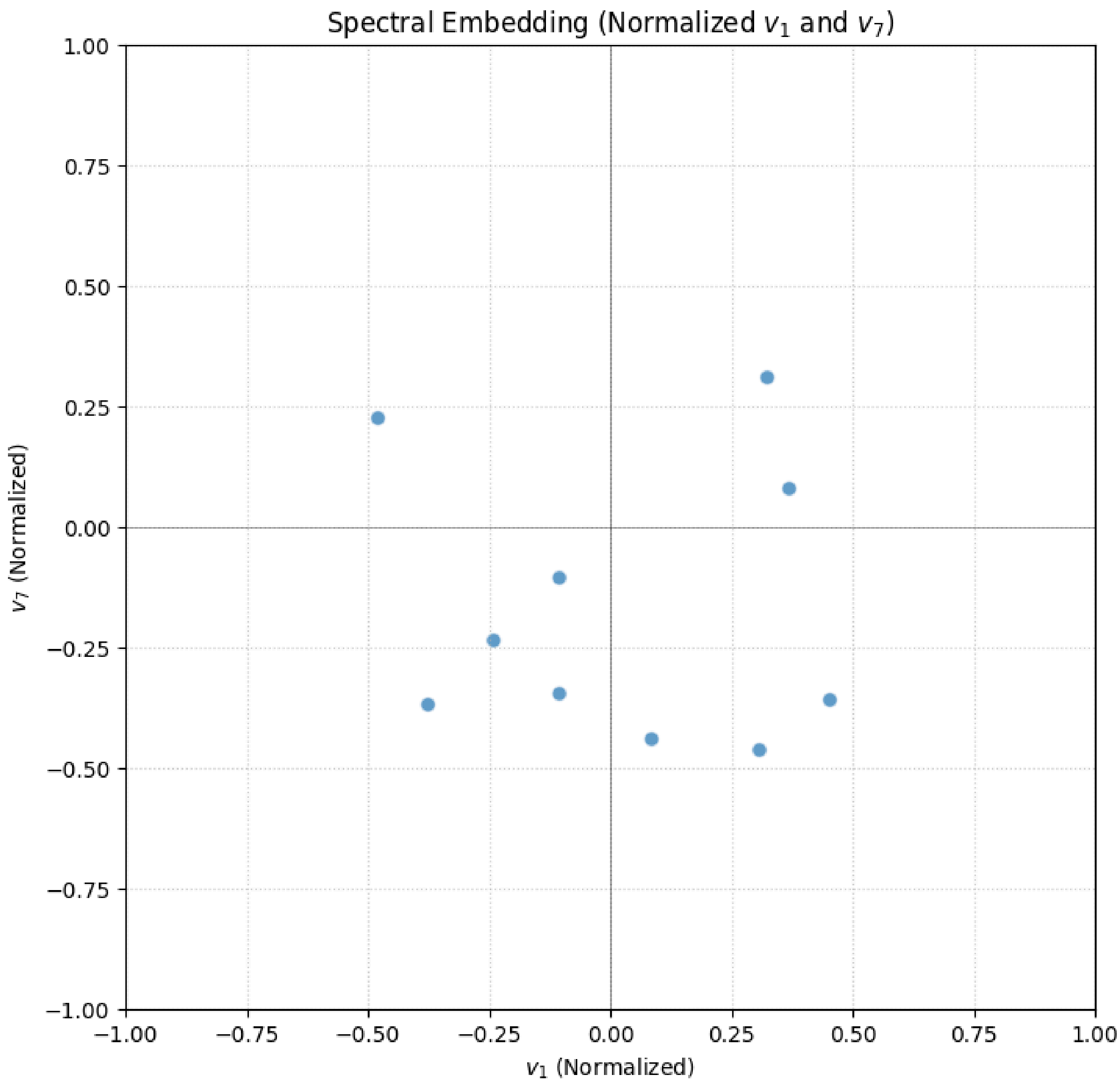
```
plot_spectral_embedding(wheel_eig[3],wheel_eig[9],3,9)
```



wheel max eigs

In [154…

```
plot_spectral_embedding(wheel_eig[1],wheel_eig[7],1,7)
```



the spectral embeddings with small eigenvectors are allowing one to see how the graph is connected. info not captured by second eig is captured partially by v3.

for large eigenvectors, the eigenvectors show how separated two vertecies are.